

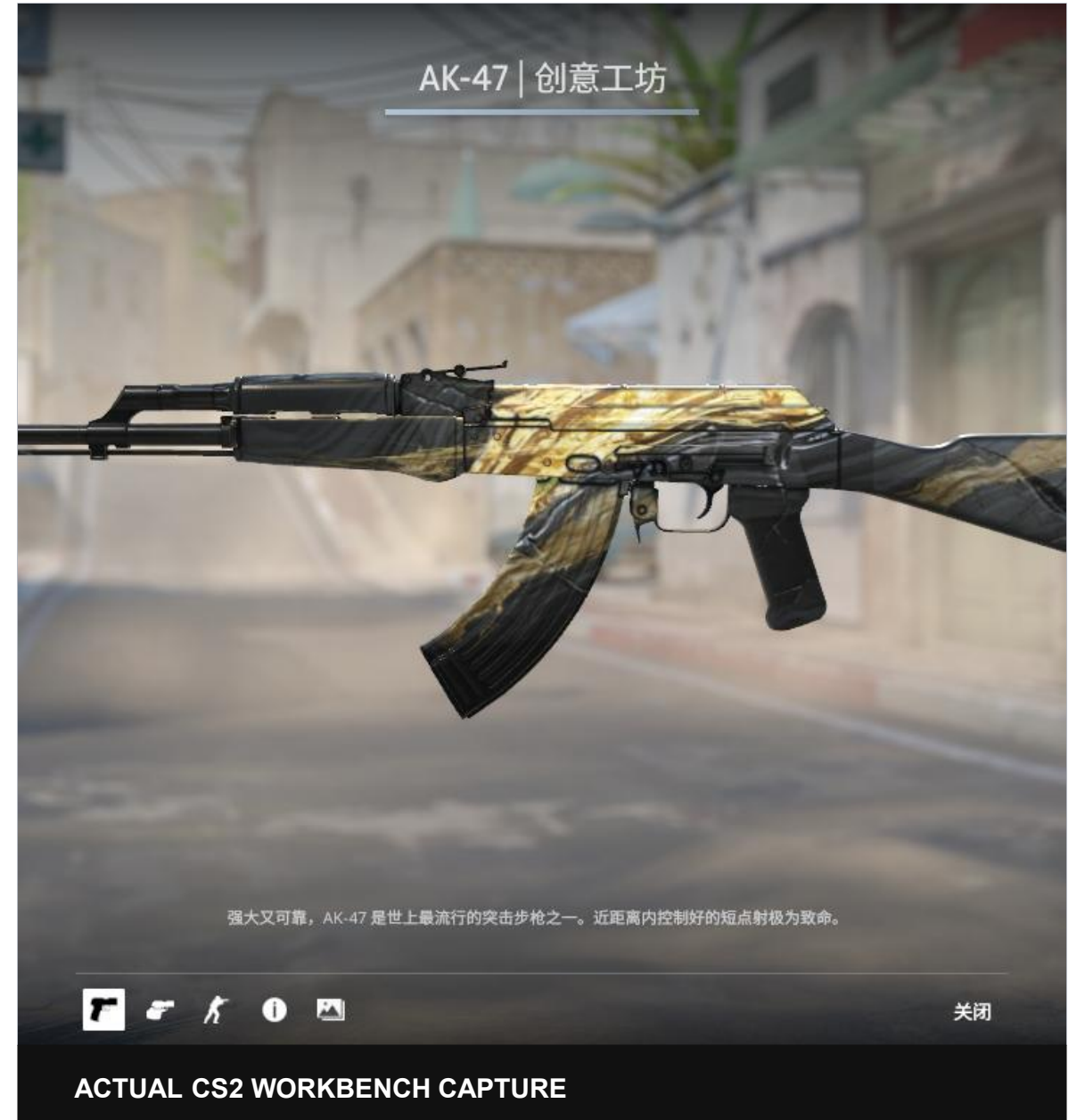
SkinSmith

A Constraint-Aware Generative Agent for Game Weapon Skin Design

A Counter-Strike 2 case study in mapped-asset evaluation

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Wang Zhengye · WANG Lihui

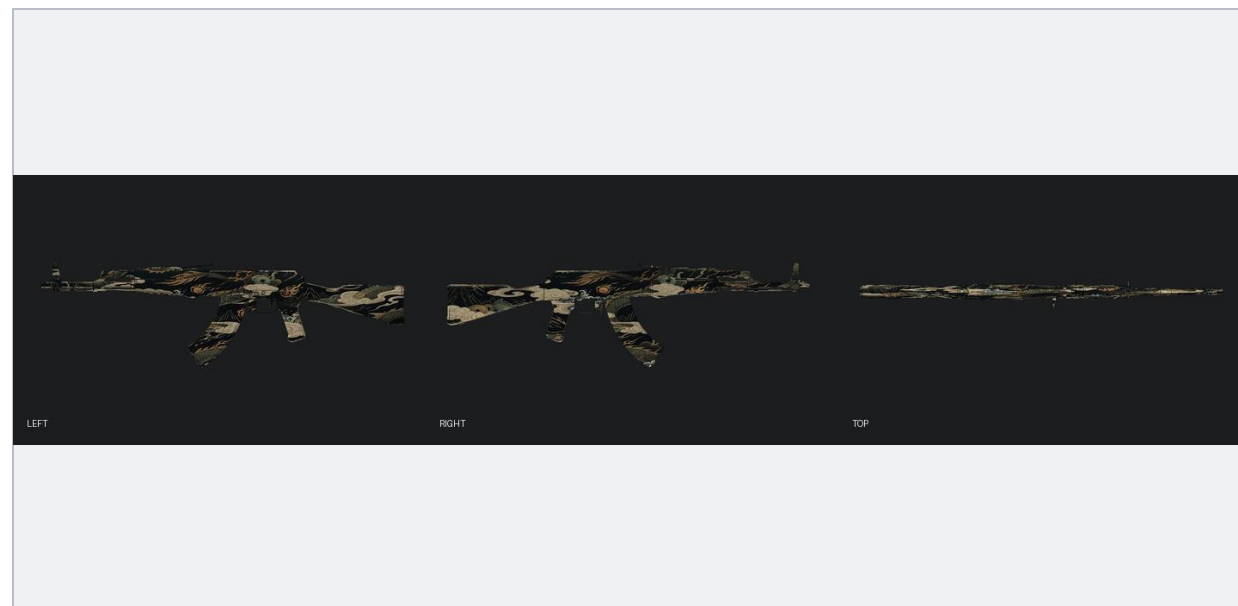
Supervisor: Prof. Yu Yang



Source-image quality is not deployment evidence



SOURCE IMAGE



RECONSTRUCTED ASSET VIEWS

UV islands split and rotate the visible surface. A motif can disappear, a seam can open, or one view can succeed while another fails.

Three checkpoints preserve user authority

01 · THEME WORLD

Confirm what “dragon” means

Agent expands subjects, symbols, materials, palette, atmosphere and exclusions.

02 · ART DIRECTION

Choose a distinct visual route

User selects one of 3–4 materially different text directions.

03 · MAPPED ARTWORK

Judge source and views together

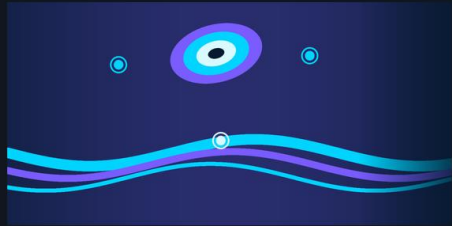
Agent maps 3–4 candidates to left, right and top views before selection.

Formal execution is blocked until Checkpoint 3 selects one indivisible source + left/right/top candidate card.

Compose continuously; evaluate after mapping

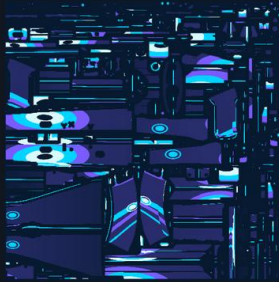
1. Weapon-space design

Coherent composition before UV packing



2. UV atlas storage

Fragments may look unrelated in 2D



3. 3D reconstruction

The model restores the intended whole



1 Weapon space

Arrange the design in a continuous canonical weapon frame.

2 Asset-aware bake

Project through OBJ positions, normals, components and UV islands.

3 Render-in-the-loop

Measure true asset seams and fixed multi-view retention.

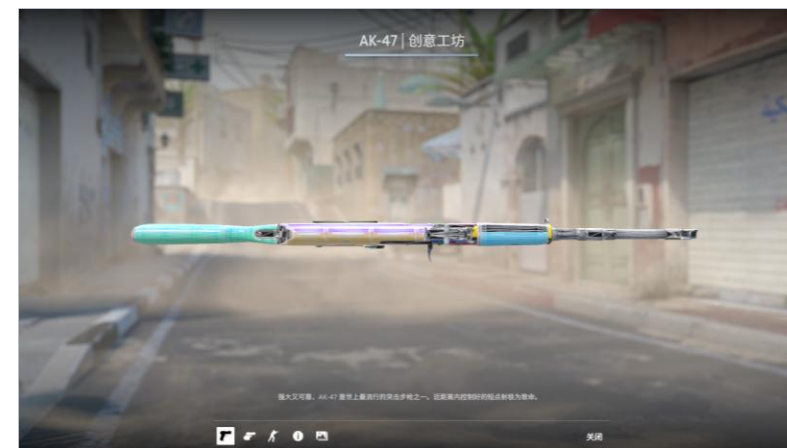
Calibration verifies the UV deployment contract



LEFT



RIGHT



TOP

0°

Rotation

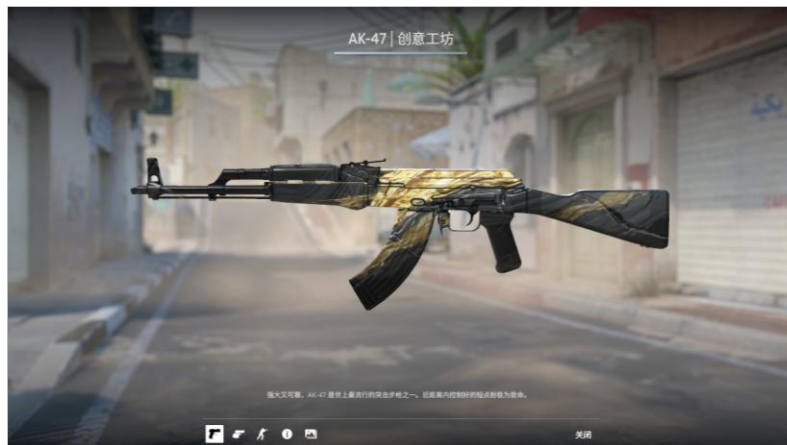
0 / 0

U / V offsets

PASS

No V flip, rotation, offset or island mismatch

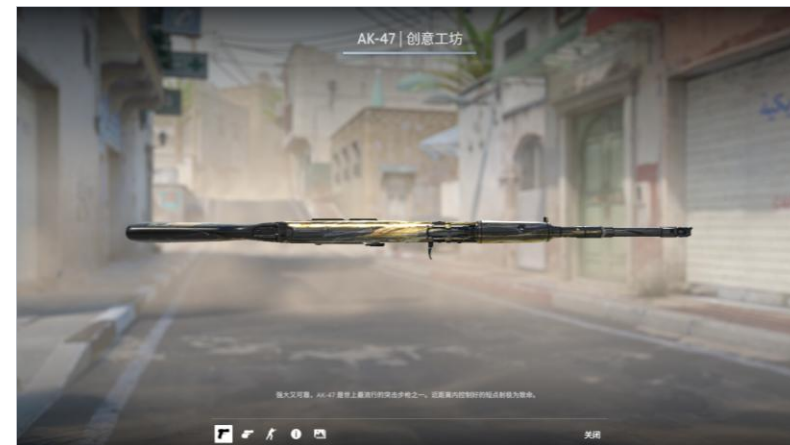
Workbench validates color-texture deployment



LEFT



RIGHT



TOP

0.000441

Corrected asset-seam error

0.749464

Corrected multi-view score

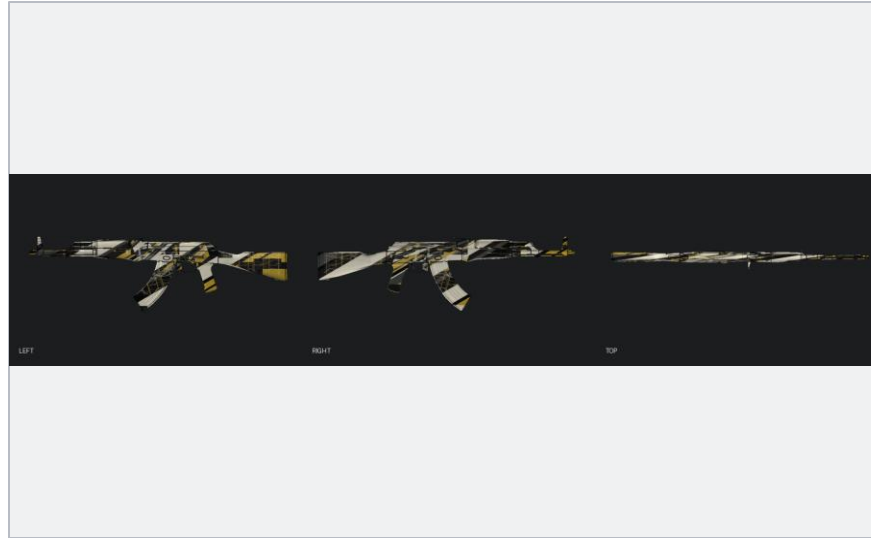
PASS

Workbench visual deployment review

Route B improves both measures in all four pairs



ROUTE A · PATTERN-FIRST BASELINE



ROUTE B · WEAPON-SPACE

99.77%

Mean seam reduction
0.26997 → 0.00063

+0.060

Mean mapped-view
score
0.76898 → 0.82911

The result is about reconstruction quality—not whether the flat UV atlas looks attractive.

One word completes all three checkpoints



dragon → Treasured Relic → artwork_04

0.000778

Asset-seam error

0.816620

Multi-view score

0.829123

Final total score

**NEXT
CAPTUR**

fixed Workbench
views

Hard gates and rollback prevent invalid choices

01

GATE

Reject candidates that violate the asset-seam threshold.

02

SELECT

Rank only feasible candidates by mapped quality and preference.

03

REFINE ONCE

Apply one local correction when the diagnosis is justified.

04

ROLL BACK

Retain Route B when correction fails the improvement rule.

Route C score change

-0.021959

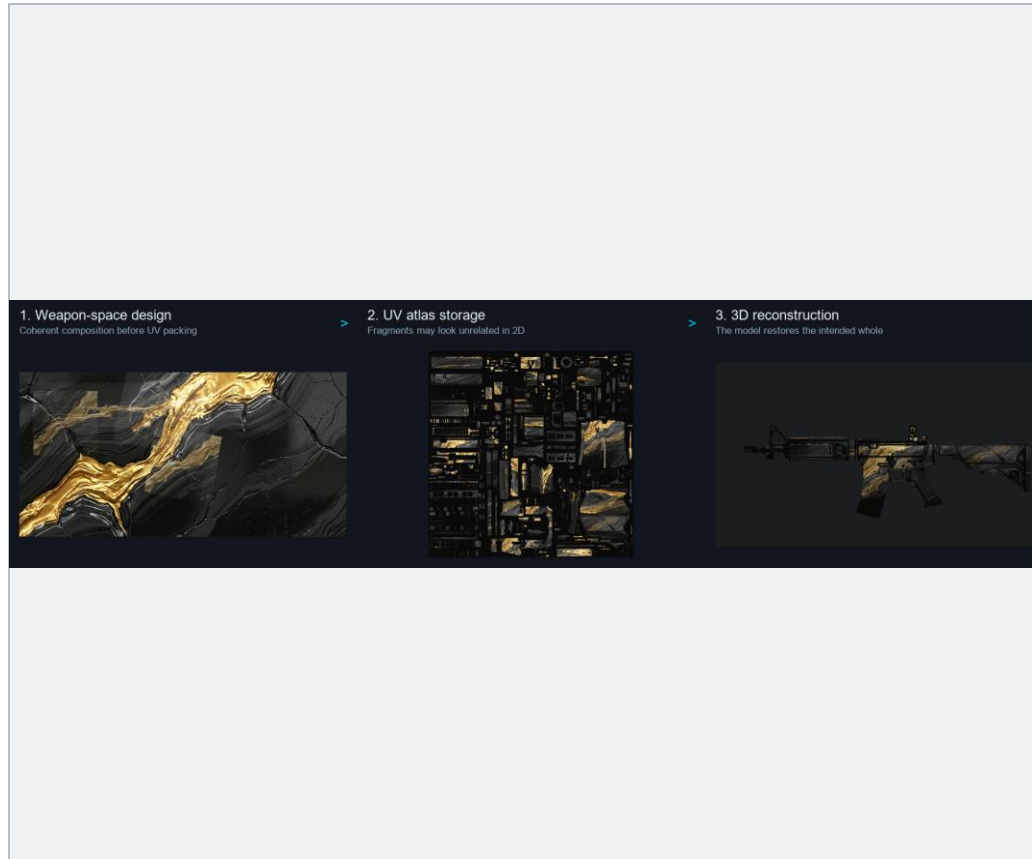
Correction rejected · Route B retained

Agent contracts transfer; UV parameters do not

AK-47

COMPLETE CASE

Three checkpoints
Formal A/B/C experiment
2048 PNG/TGA export
Workbench deployment
evidence



M4A4

TRANSFER EVIDENCE

Adapter contract
works.

Joint acceptance
threshold is not
yet met.

Not presented as
an equivalent
completed target.

Current output is color texture—not full PBR

CURRENTLY VERIFIED

Base color / custom-paint texture

- UV-aware 2048 PNG/TGA export
- Fixed left/right/top evaluation
- Asset-seam and mapped-view scoring
- Workbench color-texture deployment

NEXT MATERIAL LAYER

Normal · Roughness · Metallic

1. Generate and validate aligned material channels
2. Automate fixed-view engine capture
3. Expand asset and theme coverage
4. Run a user study of the three checkpoints

Workbench lighting must not be interpreted as model-generated surface detail.

Generation becomes an evidence-backed workflow

3

Human checkpoints separate meaning, direction and mapped choice

4 / 4

Paired candidates improve seam and mapped-view measures

1

Bounded correction with an explicit rollback decision

Main conclusion

**The generated rectangle is an intermediate artifact.
Acceptance belongs to the reconstructed asset and its preserved evidence.**

FINAL EVIDENCE TASK

**Capture final dragon
TGA in Workbench:
left, right, top +
settings.**

Questions

Five connected work packages

YUAN Ye — project lead, architecture and integration

CHEN Yuhong — creative planning and literature

Ben Tangjie — generation backends and reproducibility

Wang Zhengye — asset, UV and rendering validation

WANG Lihui — evaluation, reporting and delivery QA

github.com/DID-FIELD/SDSC6002-SkinSmith

